



DEPARTAMENTO DE CIENCIAS DE LA COMPUTACIÓN

CARRERA:

INGENIERÍA DE SOFTWARE

NRC:

2546

ASIGNATURA:

APLICACIONES DISTRIBUIDAS

DOCENTE:

DARIO MORALES

TEMA:

USO DE ORM

NOMBRES:

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FECHA:

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1. Introducción

En este informe se detalla el desarrollo de un sistema de gestión de microcursos utilizando tecnologías como Spring Boot, Docker, MySQL y Hibernate. Se presenta una visión general del proyecto, la configuración del entorno, la implementación de las funcionalidades CRUD y el despliegue mediante contenedores Docker.

2. Marco Teórico

Spring Boot

Spring Boot es un framework que facilita la creación de aplicaciones Java al proporcionar configuraciones predeterminadas y herramientas para la integración rápida de tecnologías como bases de datos, seguridad, y servicios REST. Se basa en el ecosistema de Spring Framework y es ampliamente utilizado para aplicaciones modernas.

Docker

Docker es una plataforma de contenedores que permite empaquetar aplicaciones y sus dependencias en un entorno aislado, garantizando consistencia entre entornos de desarrollo y producción.

Hibernate

Hibernate es un framework ORM (Object-Relational Mapping) que simplifica la interacción entre aplicaciones Java y bases de datos relacionales, permitiendo trabajar con objetos en lugar de sentencias SQL.

MySQL

MySQL es un sistema de gestión de bases de datos relacional de código abierto, ampliamente utilizado por su eficiencia, escalabilidad y facilidad de uso.

3. Desarrollo

Configuración del Proyecto

1. Creación del proyecto con Spring Initializr:

- Dependencias: Spring Web, Spring Data JPA, MySQL Driver, Spring Boot DevTools.
- Generación del proyecto base.

2. Estructura del proyecto:

- Paquete model.entity: Definición de las entidades (Course, Student).
- Paquete repositories: Interfaces para operaciones CRUD.
- Paquete services: Lógica de negocio.
- Paquete controllers: Endpoints REST para acceso a la API.

3. Código:

- Entidades (model.entity)

- Curso

```
6  @Entity
7  @Table(name="Cursos")
8  public class Curso {
9
10     @Id
11     @GeneratedValue(strategy = GenerationType.IDENTITY)
12     private long id;
13
14     @Column(nullable = false) 2 usages
15     private String nombre;
16
17     @Column(nullable = false) 2 usages
18     private String descripcion;
19
20     @Column(nullable = false) 2 usages
21     private String credits;
22
23     @Column(name = "fecha_creacion", updatable = false, nullable = false) 3 usages
24     private LocalDateTime fechaCreacion;
25
26     @PrePersist
27     protected void prePersist() {
28         this.fechaCreacion = LocalDateTime.now();
29     }
29 }
```

- Estudiante

```

7  @Entity
8  @Table(name = "Estudiantes")
9  public class Estudiante {
10
11      @Id
12      @GeneratedValue(strategy = GenerationType.IDENTITY)
13      private Long id;
14
15      @Column(nullable = false) 2 usages
16      private String nombre;
17
18      @Column(nullable = false) 2 usages
19      private String apellido;
20
21      @Column(nullable = false, unique = true) 2 usages
22      private String email;
23
24      @Column(name = "fecha_nacimiento", nullable = false) 2 usages
25      @Temporal(TemporalType.DATE)
26      private Date fechaNacimiento;
27
28      @Column(nullable = false) 2 usages
29      private String telefono;
30
31      @Column(name = "fecha_creacion", updatable = false, nullable = false) 3 usages
32      private LocalDateTime fechaCreacion;
33
34      @PrePersist
35      protected void prePersist() {
36          this.fechaCreacion = LocalDateTime.now();
37      }

```

- **Repositories**

- CursoRepository

```

1  package com.espe.micro_cursos.repositories;
2
3  import com.espe.micro_cursos.model.entity.Curso;
4  import org.springframework.data.repository.CrudRepository;
5
6  public interface CursoRepository extends CrudRepository<Curso, Long> { 2 usages
7

```

- EstudianteRepository

```

1  package com.espe.micro_cursos.repositories;
2
3  import com.espe.micro_cursos.model.entity.Estudiante;
4  import org.springframework.data.repository.CrudRepository;
5
6  public interface EstudianteRepository extends CrudRepository<Estudiante, Long> { 2 usages
7
8  }

```

- **Services**

- CursoService

```

1  package com.espe.micro_cursos.services;
2
3  import com.espe.micro_cursos.model.entity.Curso;
4
5  import java.util.List;
6  import java.util.Optional;
7
8  public interface CursoService { 3 usages 1 implementation
9      List<Curso> findAll(); 1 usage 1 implementation
10     Optional<Curso> findById(Long id); 3 usages 1 implementation
11     Curso save(Curso curso); 2 usages 1 implementation
12     void deleteById(Long id); 1 usage 1 implementation
13
14 }

```

- CursoServiceImpl

```
1 package com.espe.micro_cursos.services;
2
3 import com.espe.micro_cursos.model.entity.Curso;
4 import com.espe.micro_cursos.repositories.CursoRepository;
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7
8 import java.util.List;
9 import java.util.Optional;
10
11 @Service
12 public class CursoServiceImpl implements CursoService{
13     @Autowired
14     private CursoRepository repository;
15
16     @Override 1 usage
17     public List<Curso> findAll() { return (List<Curso>) repository.findAll(); }
18
19     @Override 3 usages
20     public Optional<Curso> findById(Long id) { return repository.findById(id); }
21
22     @Override 2 usages
23     public Curso save(Curso curso) { return repository.save(curso); }
24
25     @Override 1 usage
26     public void deleteById(Long id) { repository.deleteById(id); }
27 }
```

- EstudianteService

```
1 package com.espe.micro_cursos.services;
2
3 import com.espe.micro_cursos.model.entity.Estudiante;
4
5 import java.util.List;
6 import java.util.Optional;
7
8 public interface EstudianteService { 3 usages 1 implementation
9     List<Estudiante> findAll(); 1 usage 1 implementation
10     Optional<Estudiante> findById(Long id); 3 usages 1 implementation
11     Estudiante save(Estudiante estudiante); 2 usages 1 implementation
12     void deleteById(Long id); 1 usage 1 implementation
13 }
```

- EstudianteServiceImpl

```
1 import com.espe.micro_cursos.repositories.EstudianteRepository;
2 import org.springframework.beans.factory.annotation.Autowired;
3 import org.springframework.stereotype.Service;
4
5 import java.util.List;
6 import java.util.Optional;
7
8 @Service
9 public class EstudianteServiceImpl implements EstudianteService {
10     @Autowired
11     private EstudianteRepository repository;
12
13     @Override 1 usage
14     public List<Estudiante> findAll() {
15         return (List<Estudiante>) repository.findAll();
16     }
17
18     @Override 3 usages
19     public Optional<Estudiante> findById(Long id) {
20         return repository.findById(id);
21     }
22
23     @Override 2 usages
24     public Estudiante save(Estudiante estudiante) {
25         return repository.save(estudiante);
26     }
27
28     @Override 1 usage
29     public void deleteById(Long id) {
30         repository.deleteById(id);
31     }
32 }
```

- Controladores
 - CursoController

```
13 @RestController
14 @RequestMapping("/api/cursos")
15 public class CursoController {
16     @Autowired
17     private CursoService service;
18     // Crear un curso
19     @PostMapping
20     public ResponseEntity<?> crear(@RequestBody Curso curso) {
21         // La fecha de creación se asignará automáticamente en el método @PrePersist
22         Curso nuevoCurso = service.save(curso);
23         return ResponseEntity.status(HttpStatus.CREATED).body(nuevoCurso);
24     }
25     // Obtener todos los cursos
26     @GetMapping
27     public List<Curso> listar() {
28         return service.findAll();
29     }
30     // Obtener un curso por ID
31     @GetMapping("/{id}")
32     public ResponseEntity<?> buscarPorId(@PathVariable Long id) {
33         Optional<Curso> curso = service.findById(id);
34         if (curso.isPresent()) {
35             return ResponseEntity.ok(curso.get());
36         }
37         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Curso no encontrado");
38     }
39     // Actualizar un curso
40     @PutMapping("/{id}")
41     public ResponseEntity<?> actualizar(@RequestBody Curso curso, @PathVariable Long id) {
42         Optional<Curso> cursoExistente = service.findById(id);
43         if (cursoExistente.isPresent()) {
44             Curso cursoActualizado = cursoExistente.get();
45             cursoActualizado.setNombre(curso.getNombre());
46             cursoActualizado.setDescripcion(curso.getDescripcion());
47             cursoActualizado.setCreditos(curso.getCreditos());
48             // La fecha de creación no se modifica porque está configurada como "updatable = false"
49             return ResponseEntity.ok(service.save(cursoActualizado));
50         }
51         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Curso no encontrado");
52     }
53     // Eliminar un curso
54     @DeleteMapping("/{id}")
55     public ResponseEntity<?> eliminar(@PathVariable Long id) {
56         Optional<Curso> curso = service.findById(id);
57         if (curso.isPresent()) {
58             service.deleteById(id);
59             return ResponseEntity.ok("Curso eliminado correctamente");
60         }
61         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Curso no encontrado");
62     }
63 }
```

- EstudianteController

```
13 @RestController
14 @RequestMapping("/api/estudiantes")
15 public class EstudianteController {
16     @Autowired
17     private EstudianteService service;
18     // Crear un estudiante
19     @PostMapping
20     public ResponseEntity<?> crear(@RequestBody Estudiante estudiante) {
21         return ResponseEntity.status(HttpStatus.CREATED).body(service.save(estudiante));
22     }
23     // Obtener todos los estudiantes
24     @GetMapping
25     public List<Estudiante> listar() {
26         return service.findAll();
27     }
28     // Obtener un estudiante por ID
29     @GetMapping("/{id}")
30     public ResponseEntity<?> buscarPorId(@PathVariable Long id) {
31         Optional<Estudiante> estudiante = service.findById(id);
32         if (estudiante.isPresent()) {
33             return ResponseEntity.ok(estudiante.get());
34         }
35         return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Estudiante no encontrado");
36     }
37 }
```

```

42 // Actualizar un estudiante
43 @PutMapping("/{id}")
44 public ResponseEntity<?> actualizar(@RequestBody Estudiante estudiante, @PathVariable Long id) {
45     Optional<Estudiante> estudianteExistente = service.findById(id);
46     if (estudianteExistente.isPresent()) {
47         Estudiante estudianteActualizado = estudianteExistente.get();
48         estudianteActualizado.setNombre(estudiante.getNombre());
49         estudianteActualizado.setApellido(estudiante.getApellido());
50         estudianteActualizado.setEmail(estudiante.getEmail());
51         estudianteActualizado.setFechaNacimiento(estudiante.getFechaNacimiento());
52         estudianteActualizado.setTelefono(estudiante.getTelefono());
53         return ResponseEntity.ok(service.save(estudianteActualizado));
54     }
55     return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Estudiante no encontrado");
56 }
57
58 // Eliminar un estudiante
59 @DeleteMapping("/{id}")
60 public ResponseEntity<?> eliminar(@PathVariable Long id) {
61     Optional<Estudiante> estudiante = service.findById(id);
62     if (estudiante.isPresent()) {
63         service.deleteById(id);
64         return ResponseEntity.ok(body: "Estudiante eliminado correctamente");
65     }
66     return ResponseEntity.status(HttpStatus.NOT_FOUND).body("Estudiante no encontrado");
67 }
68 }

```

- Configuración de Docker
 - docker-compose.yml

```

1  version: '3.8'
2  services:
3    mysql-db:
4      image: mysql:8.0
5      container_name: mysql-container
6      environment:
7        MYSQL_ROOT_PASSWORD: root
8        MYSQL_DATABASE: sisdb2024
9      ports:
10       - "3307:3306"
11      volumes:
12       - db_data:/var/lib/mysql
13    micro-cursos:
14      build:
15        context: .
16      container_name: micro-cursos-container
17      ports:
18       - "8002:8002"
19      environment:
20        SPRING_DATASOURCE_URL: jdbc:mysql://mysql-db:3306/sisdb2024
21        SPRING_DATASOURCE_USERNAME: root
22        SPRING_DATASOURCE_PASSWORD: root
23      depends_on:
24       - mysql-db
25      volumes:
26       db_data:

```

- DOCKERFILE

```

1  # Usar una imagen base de OpenJDK
2  FROM openjdk:17-jdk-slim
3
4  # Establecer el directorio de trabajo dentro del contenedor
5  WORKDIR /app
6
7  # Copiar el archivo JAR de la aplicación al contenedor
8  COPY target/micro-cursos-0.0.1-SNAPSHOT.jar /app/micro-cursos.jar
9
10 # Exponer el puerto que usará la aplicación
11 EXPOSE 8002
12
13 # Ejecutar el archivo JAR cuando el contenedor arranque
14 ENTRYPOINT ["java", "-jar", "micro-cursos.jar"]

```

Comandos de Construcción y Despliegue

- Construir el proyecto:

mvn clean package

```
PS D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.espe:micro-cursos >-----
[INFO] Building micro-cursos 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.4.0:clean (default-clean) @ micro-cursos ---
[INFO] Deleting D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ micro-cursos ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ micro-cursos ---
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.4.2/maven-shared-utils-3.4.2.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.4.2/maven-shared-utils-3.4.2.jar (151 kB at 175 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-incremental/1.1/maven-shared-incremental-1.1.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-incremental/1.1/maven-shared-incremental-1.1.jar (1.1 kB at 1.1 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.2.0/plexus-java-1.2.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.2.0/plexus-java-1.2.0.jar (484 kB at 186 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdox/2.0.3/qdox-2.0.3.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdox/2.0.3/qdox-2.0.3.jar (123 kB at 48 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.15.0/plexus-compiler-api-2.15.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.15.0/plexus-compiler-api-2.15.0.jar (48 kB at 18 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.15.0/plexus-compiler-manager-2.15.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.15.0/plexus-compiler-manager-2.15.0.jar (48 kB at 18 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-2.15.0/plexus-compiler-2.15.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-2.15.0/plexus-compiler-2.15.0.jar (48 kB at 18 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-api/1.0.0.v20140518/aether-api-1.0.0.v20140518.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-api/1.0.0.v20140518/aether-api-1.0.0.v20140518.jar (136 kB at 31 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-util/1.0.0.v20140518/aether-util-1.0.0.v20140518.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-util/1.0.0.v20140518/aether-util-1.0.0.v20140518.jar (146 kB at 33 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-collections4-4.4.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-collections4-4.4.jar (484 kB at 186 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-compiler-api/2.15.0/plexus-compiler-api-2.15.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-compiler-api/2.15.0/plexus-compiler-api-2.15.0.jar (48 kB at 18 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-compiler-manager/2.15.0/plexus-compiler-manager-2.15.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-compiler-manager/2.15.0/plexus-compiler-manager-2.15.0.jar (48 kB at 18 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-compiler-2.15.0/plexus-compiler-2.15.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-compiler-2.15.0/plexus-compiler-2.15.0.jar (48 kB at 18 kB/s)
[INFO] Replacing main artifact D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1\target\micro-cursos-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1\target\micro-cursos-0.0.1-SNAPSHOT.jar.original
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 32.945 s
[INFO] Finished at: 2025-01-08T14:59:08-05:00
[INFO]
```

- Construir contenedores:

docker-compose build

```
PS D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1> docker-compose build
time="2025-01-08T15:00:10-05:00" level=warning msg="D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[*] Building 156.7s (10/10) FINISHED
=> [spring-boot-app internal] load build definition from Dockerfile
=> transferring dockerfile: 473B
=> [spring-boot-app internal] load metadata for docker.io/library/openjdk:17-jdk-slim
=> [spring-boot-app auth] library/openjdk:pull token for registry-1.docker.io
=> [spring-boot-app internal] load .dockerignore
=> transferring context: 2B
=> [spring-boot-app 1/3] FROM docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438e 152.7s
=> resolve docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438e5Secfa0bc126b41f68c3f62f9774 547B / 547B
=> sha256:44d3aa8d076675d49d85180b8ced9dae210fe4df4b4dbb422b9cf384e591d0 5.0s
=> sha256:6ce99fd1fe86bd02f6ad66a8e1334878528b54b5487858a76e8c88a7a27d56 187.98MB / 187.98MB
=> sha256:aaa3b3cb27e3e520b8f116863d0580c438e5Secfa0bc126b41f68c3f62f9774 547B / 547B
=> sha256:779635c9c3d23cc8dbab2d8c1ee4cf2a9202e198dfc8f4c0b279824d9b8e0f22 953B / 953B
=> sha256:37cb44321d0423bc57266a3b4f688daf90478e4cdf2d3b8091f78531053425ed 4.80kB / 4.80kB
=> sha256:1fc172e4089f03b45d412017112b110bfec42a650eddb8491ae32e3c3 31.38MB / 31.38MB
=> extracting sha256:1fe172e4089f03b45d412017112b110bfec42a650eddb8491ae32e3c3 2.5s
=> extracting sha256:44d3aa8d076675d49d85180b8ced9dae210fe4df4b4dbb422b9cf384e591d0 0.2s
=> extracting sha256:6ce99fd1fe86bd02f6ad66a8e1334878528b54b5487858a76e8c88a7a27d56 3.7s
=> [spring-boot-app internal] load build context
=> transferring context: 64.31MB
=> [spring-boot-app 2/3] WORKDIR /app
=> [spring-boot-app 3/3] COPY target/micro-cursos-0.0.1-SNAPSHOT.jar /app/micro-cursos.jar
=> exporting layers
=> writing image sha256:e96ae6edd2fd2d40942c8e394f6ee29dce9978bdfcecd32c75727ca673ac712 0.2s
=> naming to docker.io/library/micro-cursos1-spring-boot-app 0.0s
```

- Levantar servicios:

docker-compose up

```
PS D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1> docker-compose up --build
time="2025-01-08T20:23:13-05:00" level=warning msg="D:\SEMESTRE OCT24-MAR25\DISTRIBUIDAS\PARCIAL2\CLASE\micro-cursos1\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[*] Building 1.5s (10/10) FINISHED
=> [micro-cursos internal] load build definition from Dockerfile
=> transferring dockerfile: 473B
=> [micro-cursos internal] load metadata for docker.io/library/openjdk:17-jdk-slim
=> [micro-cursos auth] library/openjdk:pull token for registry-1.docker.io
=> [micro-cursos internal] load .dockerignore
=> transferring context: 2B
=> [micro-cursos 1/3] FROM docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438e5Secfa0bc126b41f68c3f62f9774 0.0s
=> resolve docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438e5Secfa0bc126b41f68c3f62f9774 547B / 547B
=> sha256:44d3aa8d076675d49d85180b8ced9dae210fe4df4b4dbb422b9cf384e591d0 5.0s
=> sha256:6ce99fd1fe86bd02f6ad66a8e1334878528b54b5487858a76e8c88a7a27d56 187.98MB / 187.98MB
=> sha256:aaa3b3cb27e3e520b8f116863d0580c438e5Secfa0bc126b41f68c3f62f9774 547B / 547B
=> sha256:779635c9c3d23cc8dbab2d8c1ee4cf2a9202e198dfc8f4c0b279824d9b8e0f22 953B / 953B
=> sha256:37cb44321d0423bc57266a3b4f688daf90478e4cdf2d3b8091f78531053425ed 4.80kB / 4.80kB
=> sha256:1fc172e4089f03b45d412017112b110bfec42a650eddb8491ae32e3c3 31.38MB / 31.38MB
=> extracting sha256:1fe172e4089f03b45d412017112b110bfec42a650eddb8491ae32e3c3 2.5s
=> extracting sha256:44d3aa8d076675d49d85180b8ced9dae210fe4df4b4dbb422b9cf384e591d0 0.2s
=> extracting sha256:6ce99fd1fe86bd02f6ad66a8e1334878528b54b5487858a76e8c88a7a27d56 3.7s
=> [micro-cursos internal] load build context
=> transferring context: 64.31MB
=> [micro-cursos 2/3] WORKDIR /app
=> [micro-cursos 3/3] COPY target/micro-cursos-0.0.1-SNAPSHOT.jar /app/micro-cursos.jar
=> exporting layers
=> writing image sha256:58640d6a105005a216a001ceddbbcfb8499304a3731f1bf4a258ac446571d899 0.0s
=> naming to docker.io/library/micro-cursos1-micro-cursos 0.0s
=> [micro-cursos] resolving provenance for metadata file 0.0s
[*] Running 3/2
✓ Network micro-cursos1_default Created 0.1s
✓ Container mysql-container Created 0.1s
✓ Container micro-cursos-container Created 0.1s
attaching to micro-cursos-container, mysql-container
```

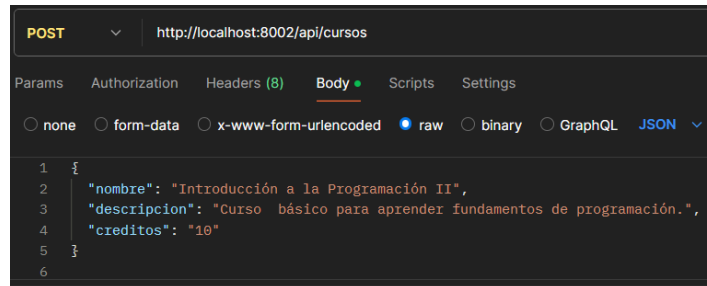
4. Pruebas

Las pruebas del sistema se realizaron utilizando Postman para verificar los endpoints CRUD. A continuación, se listan los pasos y capturas:

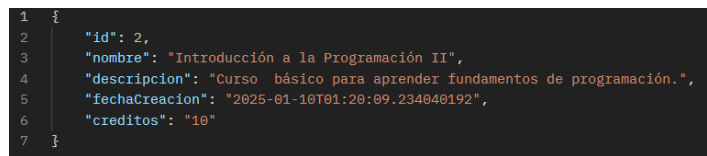
- **Cursos**

1. Crear Curso:

- Endpoint: POST /api/cursos
- Body:

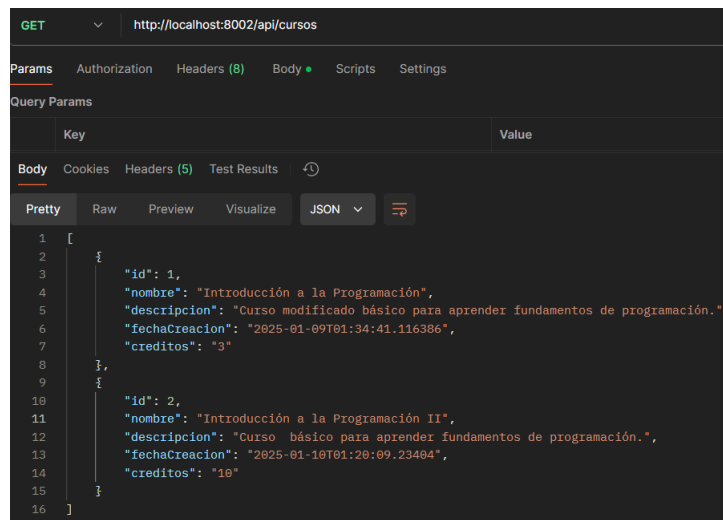


- Resultado:



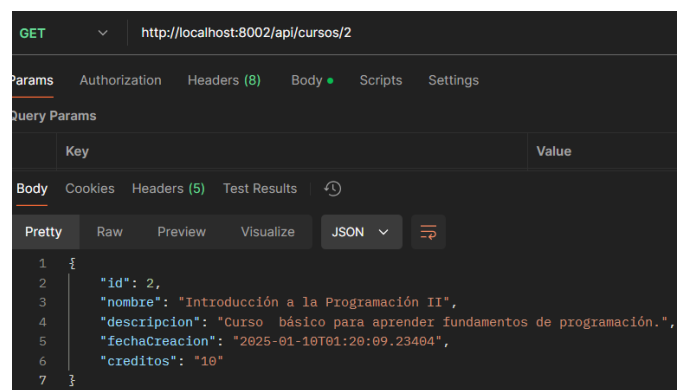
2. Listar Cursos:

- Endpoint: GET /api/cursos



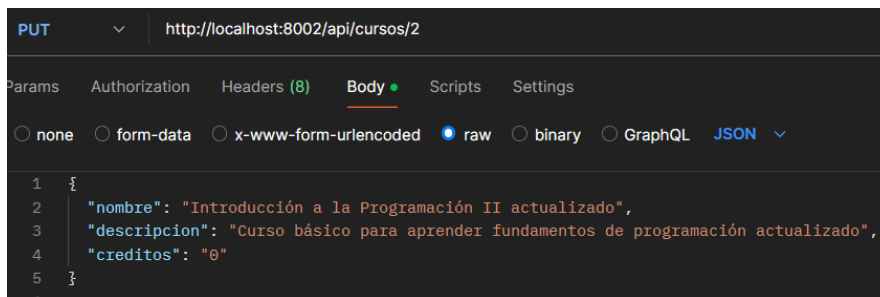
3. Listar Cursos por ID:

- Endpoint: GET /api/cursos/{id}

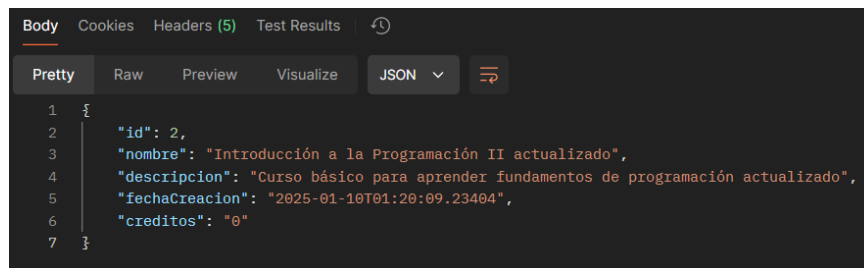


4. Actualizar Curso por ID:

- Endpoint: PUT /api/cursos/{id}
- Body:

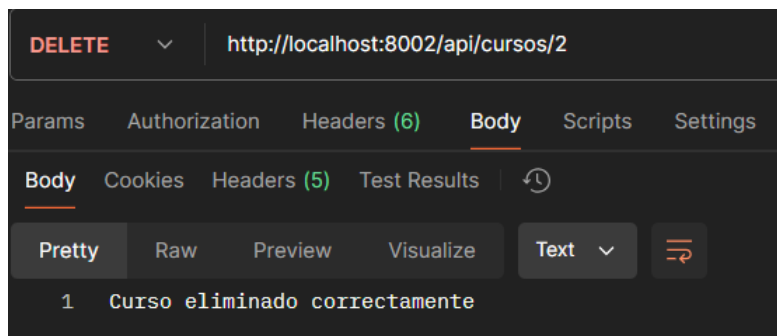


- Resultado:



5. Eliminar Curso por ID:

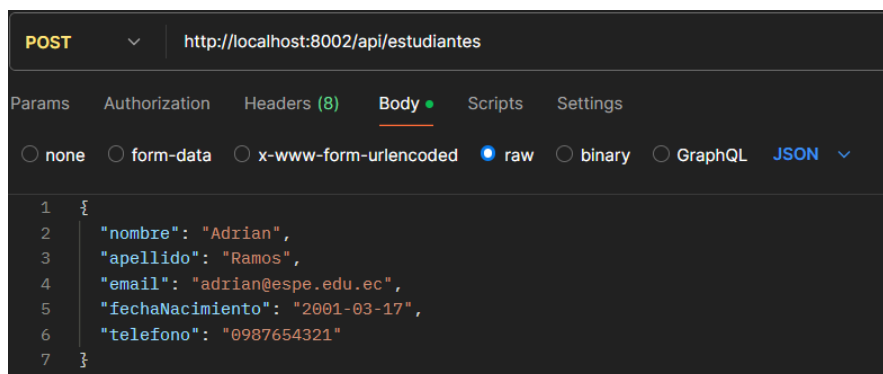
- Endpoint: DELETE /api/cursos/{id}



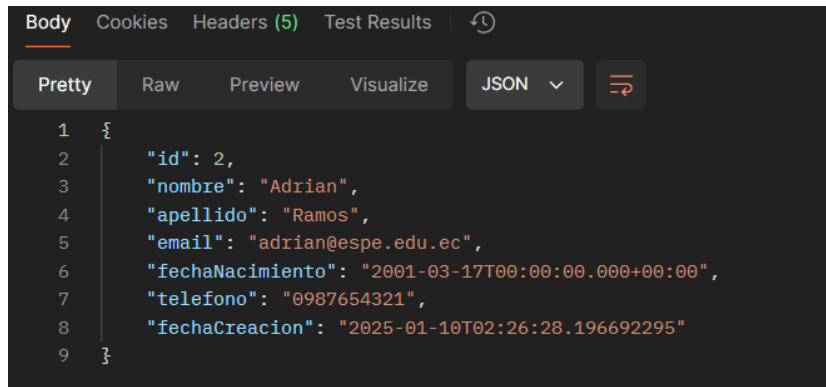
- Estudiantes

1. Crear Curso:

- Endpoint: POST /api/estudiantes
- Body:



- Resultado:

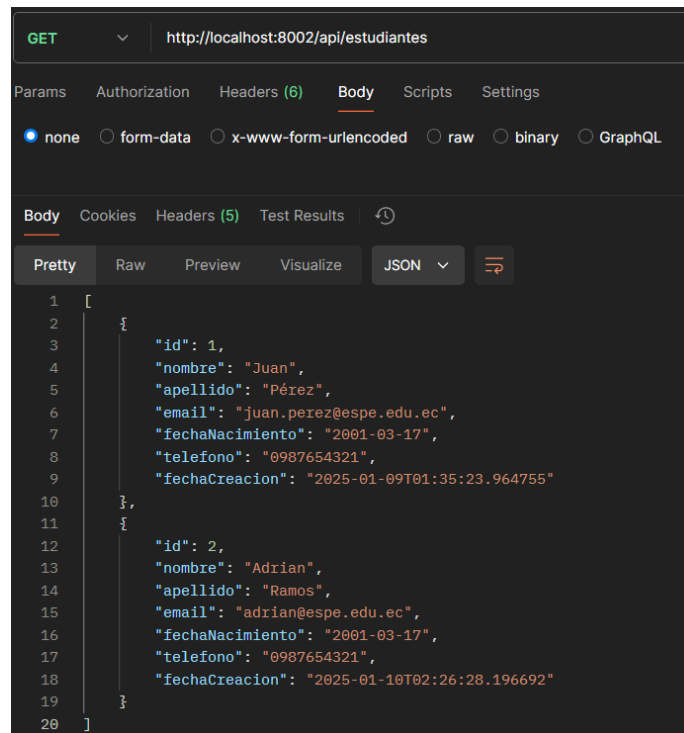


A screenshot of a REST client interface. The top bar shows tabs for 'Body', 'Cookies', 'Headers (5)', and 'Test Results'. The 'Body' tab is active, and the 'Pretty' button is selected. The response is a single JSON object with the following fields: 'id' (2), 'nombre' ('Adrian'), 'apellido' ('Ramos'), 'email' ('adrian@espe.edu.ec'), 'fechaNacimiento' ('2001-03-17T00:00:00.000+00:00'), 'telefono' ('0987654321'), and 'fechaCreacion' ('2025-01-10T02:26:28.196692295').

```
1 {
2   "id": 2,
3   "nombre": "Adrian",
4   "apellido": "Ramos",
5   "email": "adrian@espe.edu.ec",
6   "fechaNacimiento": "2001-03-17T00:00:00.000+00:00",
7   "telefono": "0987654321",
8   "fechaCreacion": "2025-01-10T02:26:28.196692295"
9 }
```

2. Listar Cursos:

- Endpoint: GET /api/estudiantes

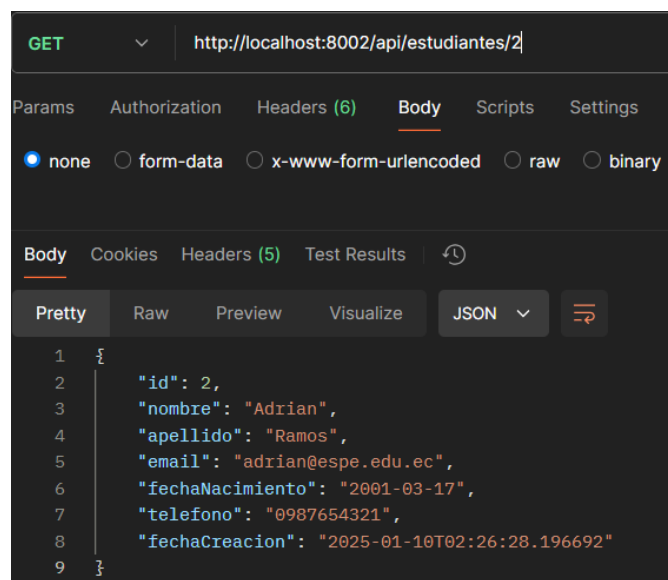


A screenshot of a REST client interface. The top bar shows the method 'GET' and the URL 'http://localhost:8002/api/estudiantes'. The 'Body' tab is active, and the 'Pretty' button is selected. The response is a JSON array containing two objects, each with the same fields as the previous response: 'id', 'nombre', 'apellido', 'email', 'fechaNacimiento', 'telefono', and 'fechaCreacion'.

```
1 [
2   {
3     "id": 1,
4     "nombre": "Juan",
5     "apellido": "Pérez",
6     "email": "juan.perez@espe.edu.ec",
7     "fechaNacimiento": "2001-03-17",
8     "telefono": "0987654321",
9     "fechaCreacion": "2025-01-09T01:35:23.964755"
10  },
11  {
12    "id": 2,
13    "nombre": "Adrian",
14    "apellido": "Ramos",
15    "email": "adrian@espe.edu.ec",
16    "fechaNacimiento": "2001-03-17",
17    "telefono": "0987654321",
18    "fechaCreacion": "2025-01-10T02:26:28.196692"
19  }
20 ]
```

3. Listar Cursos por ID:

- Endpoint: GET /api/estudiantes/{id}

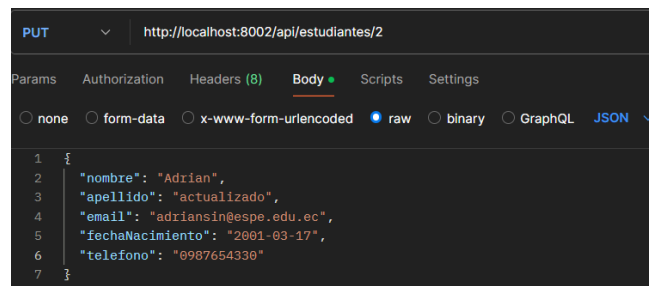


A screenshot of a REST client interface. The top bar shows the method 'GET' and the URL 'http://localhost:8002/api/estudiantes/2'. The 'Body' tab is active, and the 'Pretty' button is selected. The response is a single JSON object for the student with ID 2, containing the same fields as the previous responses.

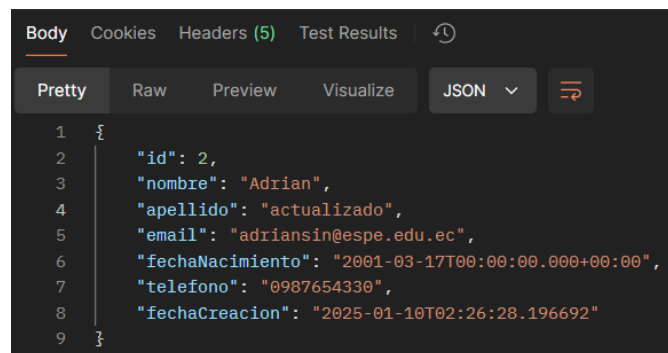
```
1 {
2   "id": 2,
3   "nombre": "Adrian",
4   "apellido": "Ramos",
5   "email": "adrian@espe.edu.ec",
6   "fechaNacimiento": "2001-03-17",
7   "telefono": "0987654321",
8   "fechaCreacion": "2025-01-10T02:26:28.196692"
9 }
```

4. Actualizar Curso por ID:

- Endpoint: PUT /api/estudiantes/{id}
- Body:

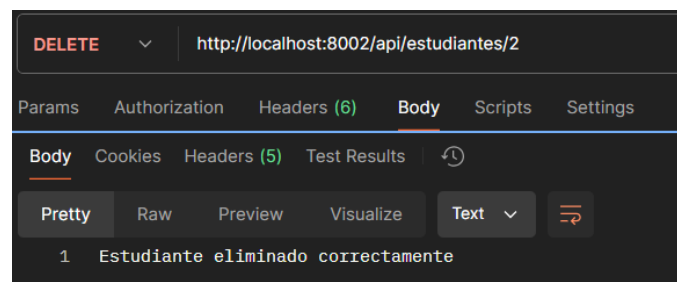


- Resultado:

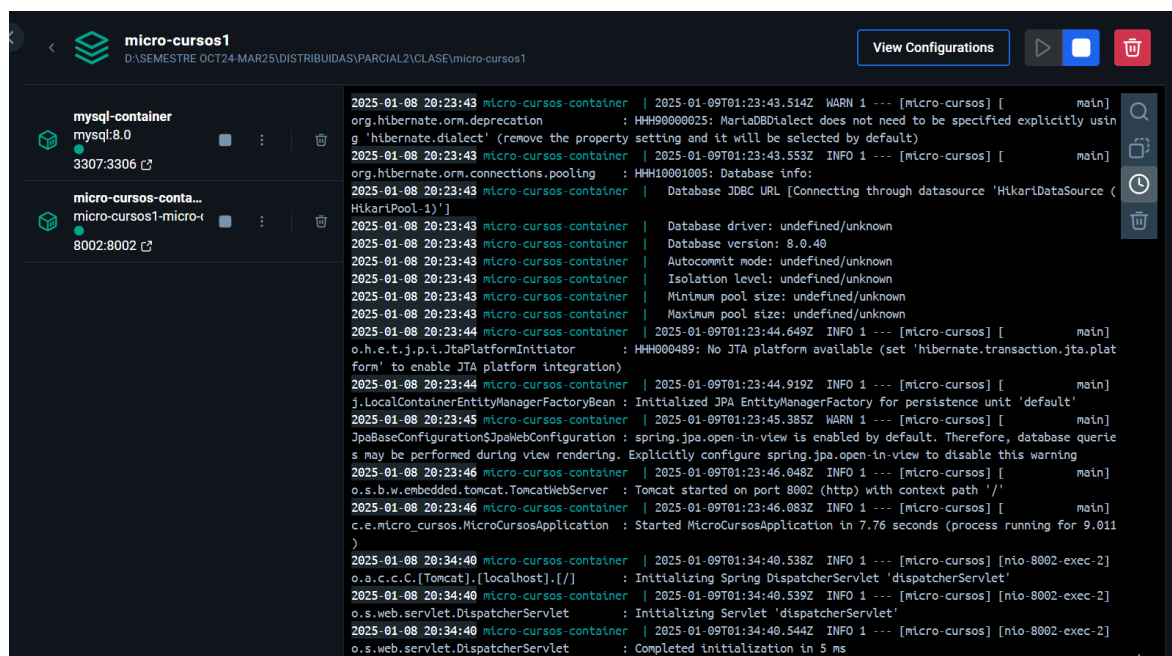


5. Eliminar Curso por ID:

- Endpoint: DELETE /api/estudiantes/{id}



- Contenedores levantados



5. Conclusión

El proyecto logró implementar un sistema funcional de gestión de microcursos con capacidades CRUD. La integración de Docker permitió un despliegue eficiente y reproducible. Las pruebas confirmaron la funcionalidad y confiabilidad del sistema.

6. Bibliografía

- Walls, C. (2019). Spring Boot in Action. Manning Publications.
- Turnbull, J. (2014). The Docker Book: Containerization is the New Virtualization.
- MySQL Documentation: <https://dev.mysql.com/doc/>
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